



ICT306 Advanced Cybersecurity

WEEK 1 Introduction to Ethical Hacking

Resources

Prescribed Reading

Meredith, D. (2022). *Certified Ethical Hacker (CEH) v12 312-50 Exam Guide: Keep up to date with ethical hacking trends and hone your skills with hands-on activities*. Packt Publishing Ltd.

Recommended Reading

Whitman, Michael E., and Herbert J. Mattord. *Principles of Information Security*. Cengage Learning, 2021.

- 📄 **Module 3** — This lecture aligns with Module 3 of the course. Ensure you complete the prescribed reading before the next class to reinforce the concepts introduced today.

Learning Outcomes

By the end of this lecture, students should be able to:

01

Define

Key concepts of hacking, ethical hacking, and cybercrime.

03

Explain

The legal and ethical responsibilities associated with hacking and cybersecurity.

05

Identify

Tools, methods, and types of penetration testing used in real-world security assessments.

02

Differentiate

Between types of hackers, attacks, and cybercrimes.

04

Describe

The phases and purposes of vulnerability assessment and penetration testing.

06

Recognize

The importance of ethical conduct and compliance with cyber laws in security operations.

Introduction to Cybercrime

Criminal Activities Targeting or Using Computer Systems and Networks

- **Cybercrime** encompasses **any illegal act perpetrated through digital means**.
- The following are the primary categories of offenses against computer data and systems:

Illegal Access (Hacking)

Unauthorized entry into computer systems or networks to steal, modify, or delete data.

Illegal Interception

Capturing or monitoring network traffic or communications, e.g., packet sniffing and eavesdropping.

Data Interference

Unauthorized alteration, deletion, or corruption of digital data or databases.

System Interference

Disruption of computer functionality, such as Denial-of-Service (DoS) attacks or malware deployment.

Misuse of Devices

Creation, possession, or distribution of tools designed to commit cybercrimes, such as keyloggers and exploit kits.



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Types of Cybercrime

- **Cybercrime takes many forms**, ranging from **individual-level attacks** to **large-scale disruptions** of national infrastructure.
- Understanding these categories helps security professionals anticipate and counter threats effectively.

Hacking

Unauthorized access to computer systems or networks to steal, modify, or destroy data.

DoS Attack

Overloading a system or network to make it unavailable to legitimate users.

Malware Dissemination

Developing or spreading malicious software to damage or gain control over systems.

Computer Vandalism

Intentional destruction, defacement, or alteration of digital content or systems.

Cyber Terrorism

Using digital attacks to disrupt critical infrastructure, create fear, or cause widespread harm.

Software Piracy

Unauthorized duplication, distribution, or use of copyrighted software.

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Motivations of Attacks

Attacks = Motive (Goal) + Method + Vulnerability

- A motive originates from the belief that **a target system stores or processes something valuable**.
- Attackers **exploit vulnerabilities** using various tools and techniques to achieve their goals.
- **Understanding motive** is key to anticipating and preventing attacks.

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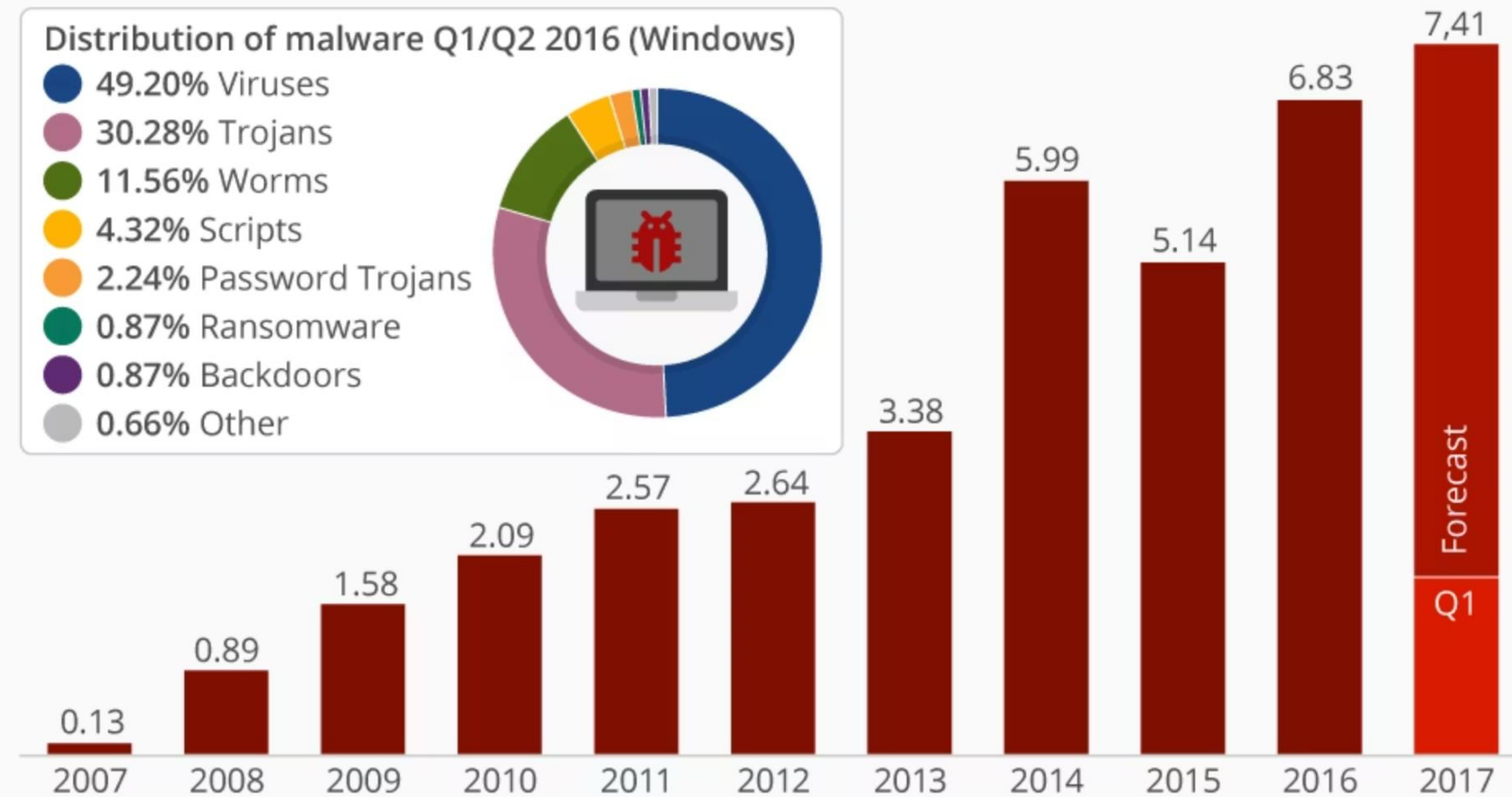
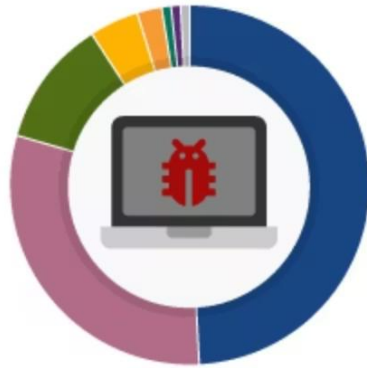
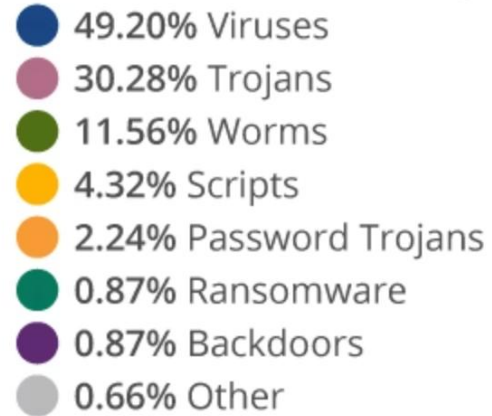
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Viruses, Worms and Trojan Horses

Number of new malware specimen (in millions)

Distribution of malware Q1/Q2 2016 (Windows)



@StatistaCharts Source: G DATA, AV-TEST

statista

Distributed Attacks: Viruses, Worms & Trojans

- **Distributed attacks** leverage malicious software spread across millions of systems worldwide.
- **Viruses, worms, and Trojans** remain the most prevalent categories of malware.
- The chart below illustrates **the explosive growth in new malware specimens** from 2007 to 2017, with viruses and Trojans dominating the threat landscape.

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Real-World Incident: The Home Depot Breach

One of the most significant retail data breaches in history, the **Home Depot incident demonstrates the devastating impact** of targeted malware attacks on large enterprises and their customers.

56M

Cards Stolen

Debit and credit card numbers compromised in the attack.

2014

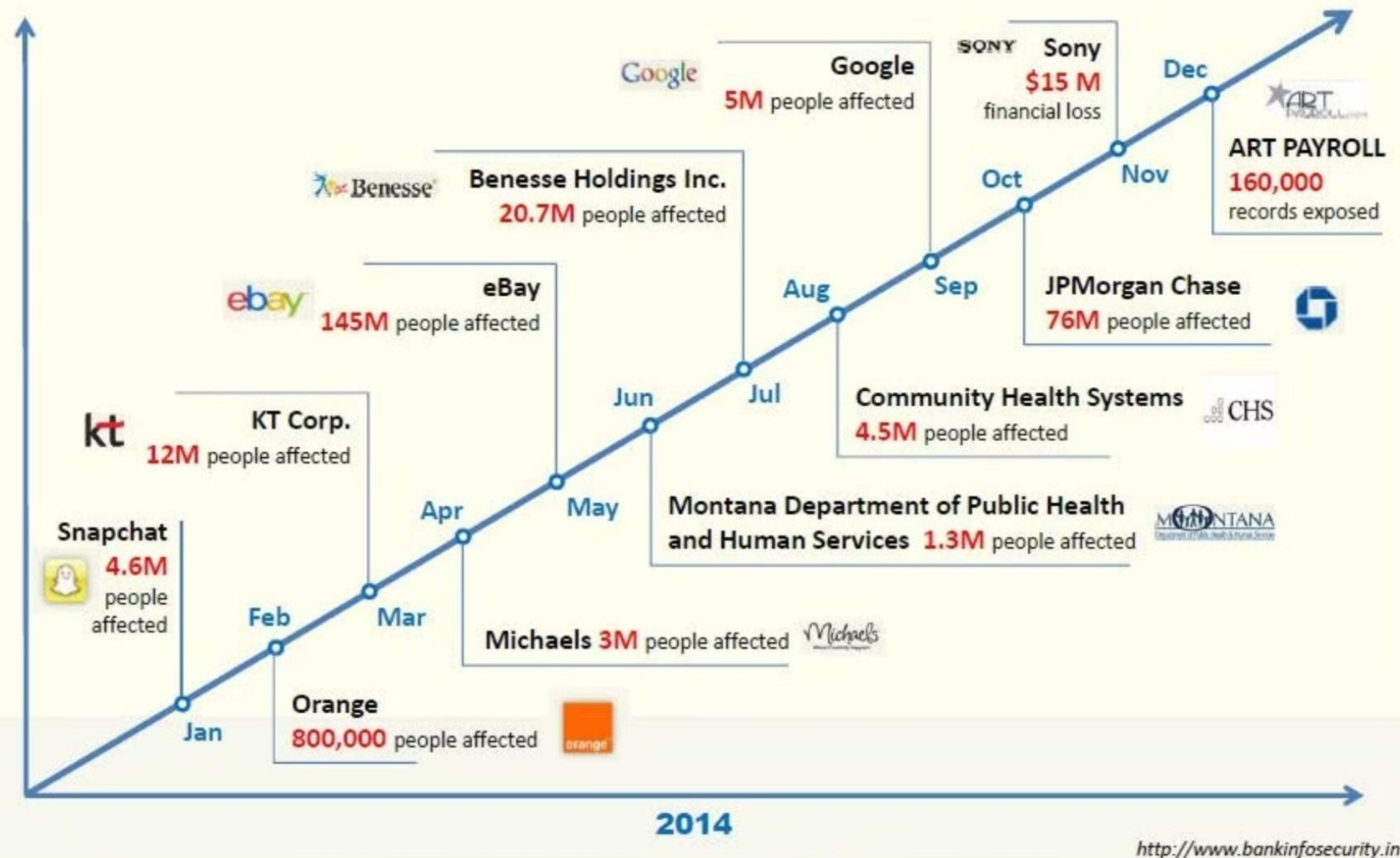
Year of Breach

Custom-built malware was used to infiltrate point-of-sale systems.



Major Cybersecurity Incidents: 2014 Timeline

- The year 2014 was marked by an unprecedented wave of high-profile data breaches affecting millions of individuals globally.
- From **Snapchat** and **eBay** to **JPMorgan Chase** and **Google**, no sector was immune.
- This timeline illustrates the scale and frequency of attacks during this critical year in cybersecurity history.

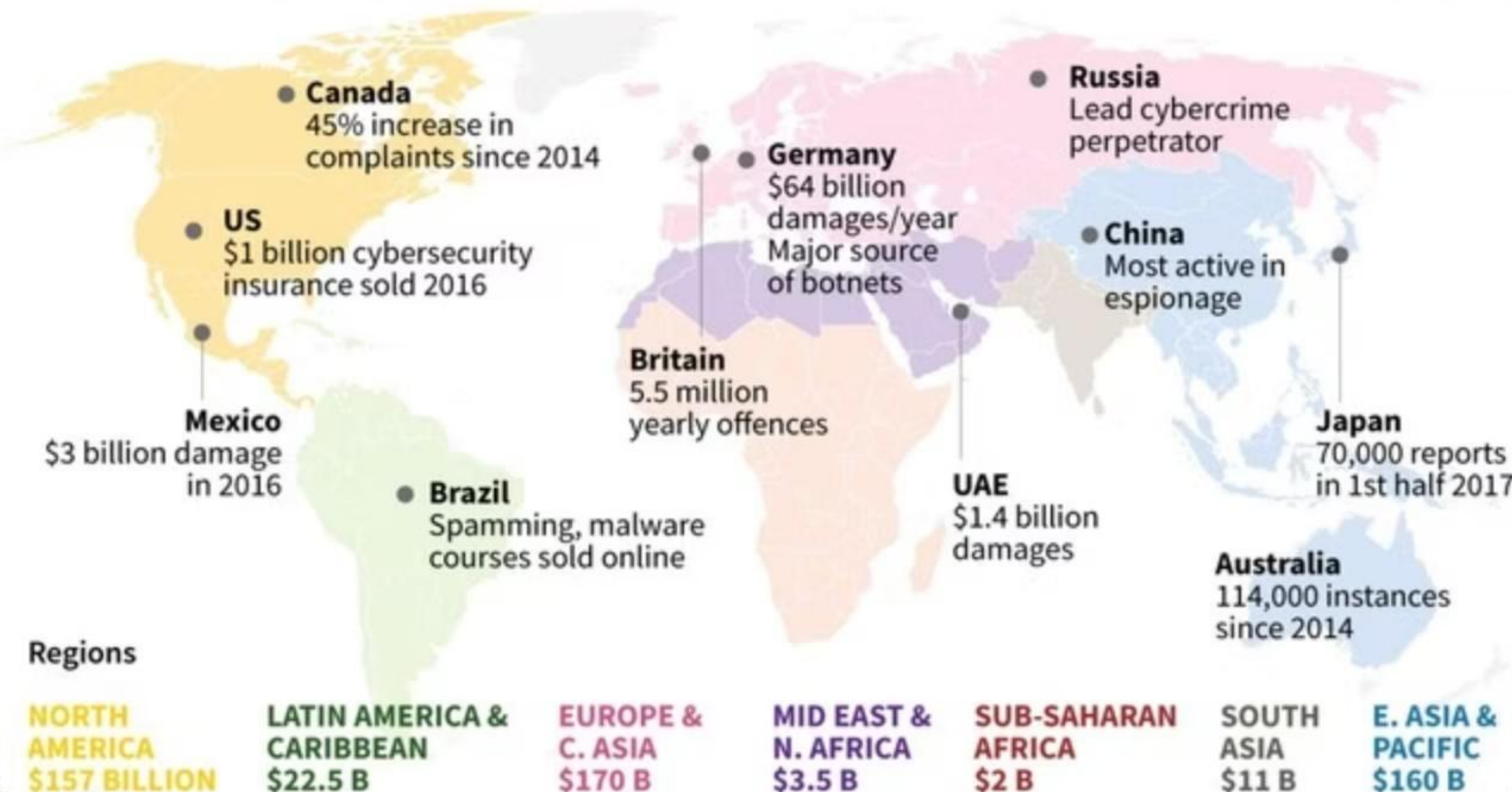


Cybercrime highlights

Economic impact: McAfee report, February 2018

Estimated annual global cost: \$600 billion

Selected highlights



Cybercrime: The Global Cost

- Cybercrime imposes a **staggering financial burden** on the global economy each year.
- From **direct theft and fraud** to the costs of remediation, lost productivity, and reputational damage, the economic impact of cyberattacks continues to escalate.
- **Organizations of every size and sector are affected**, underscoring the critical importance of proactive cybersecurity investment.

Cyber Law: The Legal Framework

- Cyber law defines the legal framework governing the use of the internet, digital communication, and information systems.
- It addresses crimes, privacy, and online conduct in the digital environment.
- Key characteristics of cyber law include:



Very Strict Regulations

Enforces strong penalties for cybercrimes and unauthorized digital activities, acting as a powerful deterrent.



Borderless Jurisdiction

Cyber offenses can be investigated and prosecuted across national boundaries, enabling international cooperation.



Arrest Without Warrant

Law enforcement may detain suspects involved in serious cyber offenses immediately to prevent further harm.



No Witness Required

Digital evidence and electronic records are often sufficient for successful prosecution in court.

Cyber Law — International Examples

- Different countries have enacted **specific legislation** to combat cybercrime and regulate digital conduct.
- These laws **establish the boundaries** of lawful behavior online and define consequences for violations.
- The table highlights **key international cyber law frameworks**.

Country Name	Laws/Acts	Website
United States	Section 107 of the Copyright Law mentions the doctrine of “fair use”	http://www.copyright.gov
	Online Copyright Infringement Liability Limitation Act	
	The Lanham (Trademark) Act (15 USC §§ 1051 - 1127)	http://www.uspto.gov
	The Electronic Communications Privacy Act	https://www.fas.org
	Foreign Intelligence Surveillance Act	https://www.fas.org
	Protect America Act of 2007	http://www.justice.gov
	Privacy Act of 1974	http://www.justice.gov
	National Information Infrastructure Protection Act of 1996	http://www.nrotc.navy.mil
	Computer Security Act of 1987	http://csrc.nist.gov
	Freedom of Information Act (FOIA)	http://www.foia.gov
	Computer Fraud and Abuse Act	http://energy.gov
	Federal Identity Theft and Assumption Deterrence Act	http://www.ftc.gov

Cyber Law — Enforcement Mechanisms

- Cyber law enforcement relies on specialized agencies, digital forensics, and international treaties.
- Effective enforcement **requires rapid response capabilities, cross-border cooperation, and technical expertise** to trace and prosecute cybercriminals operating in any jurisdiction.

Country Name	Laws/Acts	Website
Australia	The Trade Marks Act 1995	http://www.comlaw.gov.au
	The Patents Act 1990	
	The Copyright Act 1968	
	Cybercrime Act 2001	
United Kingdom	The Copyright, Etc. and Trademarks (Offenses And Enforcement) Act 2002	http://www.legislation.gov.uk
	Trademarks Act 1994 (TMA)	
	Computer Misuse Act 1990	
China	Copyright Law of People's Republic of China (Amendments on October 27, 2001)	http://www.npc.gov.cn
	Trademark Law of the People's Republic of China (Amendments on October 27, 2001)	http://www.saic.gov.cn
India	The Patents (Amendment) Act, 1999, Trade Marks Act, 1999, The Copyright Act, 1957	http://www.ipindia.nic.in
	Information Technology Act	http://www.dot.gov.in
Germany	Section 202a. Data Espionage, Section 303a. Alteration of Data, Section 303b. Computer Sabotage	http://www.cybercrimelaw.net

Cyber Law — Compliance and Responsibilities

- **Both individuals and organizations** carry legal responsibilities under cyber law.
- **Compliance requires** understanding applicable regulations, implementing adequate security controls, and maintaining transparent data handling practices.
- **Non-compliance** can result in significant fines, reputational damage, and criminal liability.

Country Name	Laws/Acts	Website
Italy	Penal Code Article 615 ter	http://www.cybercrimelaw.net
Japan	The Trademark Law (Law No. 127 of 1957), Copyright Management Business Law (4.2.2.3 of 2000)	http://www.iip.or.jp
Canada	Copyright Act (R.S.C., 1985, c. C-42), Trademark Law, Canadian Criminal Code Section 342.1	http://www.laws-lois.justice.gc.ca
Singapore	Computer Misuse Act	http://www.statutes.agc.gov.sg
South Africa	Trademarks Act 194 of 1993	http://www.cipc.co.za
	Copyright Act of 1978	http://www.nlsa.ac.za
South Korea	Copyright Law Act No. 3916	http://home.heinonline.org
	Industrial Design Protection Act	http://www.kipo.go.kr
Belgium	Copyright Law, 30/06/1994	http://www.wipo.int
	Computer Hacking	http://www.cybercrimelaw.net
Brazil	Unauthorized modification or alteration of the information system	http://www.mosstingrett.no
Hong Kong	Article 139 of the Basic Law	http://www.basiclaw.gov.hk

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What Is Hacking?

Hacking refers to the practice of exploiting weaknesses in computer systems, networks, or applications to gain unauthorized access or achieve unintended outcomes.

While often portrayed negatively, hacking encompasses a broad spectrum of activities — from malicious intrusion to legitimate security research.

Types of Hackers

- Not all hackers share the same motives or methods.
- The cybersecurity community categorizes hackers based on their intent, authorization, and the nature of their activities.
- Understanding these distinctions is fundamental for any security professional.

White Hat

Authorized ethical hackers who test systems with permission to improve security.

Black Hat

Malicious attackers who breach systems without authorization for personal gain or harm.

Gray Hat

Operate between ethical and unethical boundaries — may expose flaws without prior permission.



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Ethical Hacking: An Overview

Ethical hacking — also known as penetration testing or white-hat hacking — involves the authorized simulation of cyberattacks to identify and remediate vulnerabilities before malicious actors can exploit them. It is a critical discipline within modern cybersecurity.

Ethical Hacking: Core Principles

Ethical hacking is governed by strict principles that distinguish it from malicious hacking. Adherence to these principles ensures that security testing is conducted responsibly, legally, and with the organization's best interests in mind.

Ethical Hacking: Methodology

Ethical hackers follow a structured methodology to systematically uncover and report vulnerabilities. This disciplined approach ensures comprehensive coverage of the attack surface and produces actionable findings that organizations can use to strengthen their defenses.

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Skills of a Hacker

- Both **ethical** and **malicious hackers** rely on a **diverse and deep skill set**.
- Understanding what skills attackers possess helps defenders anticipate threats and build stronger security programs.
- Proficiency in these areas is also essential for aspiring ethical hackers and security professionals.

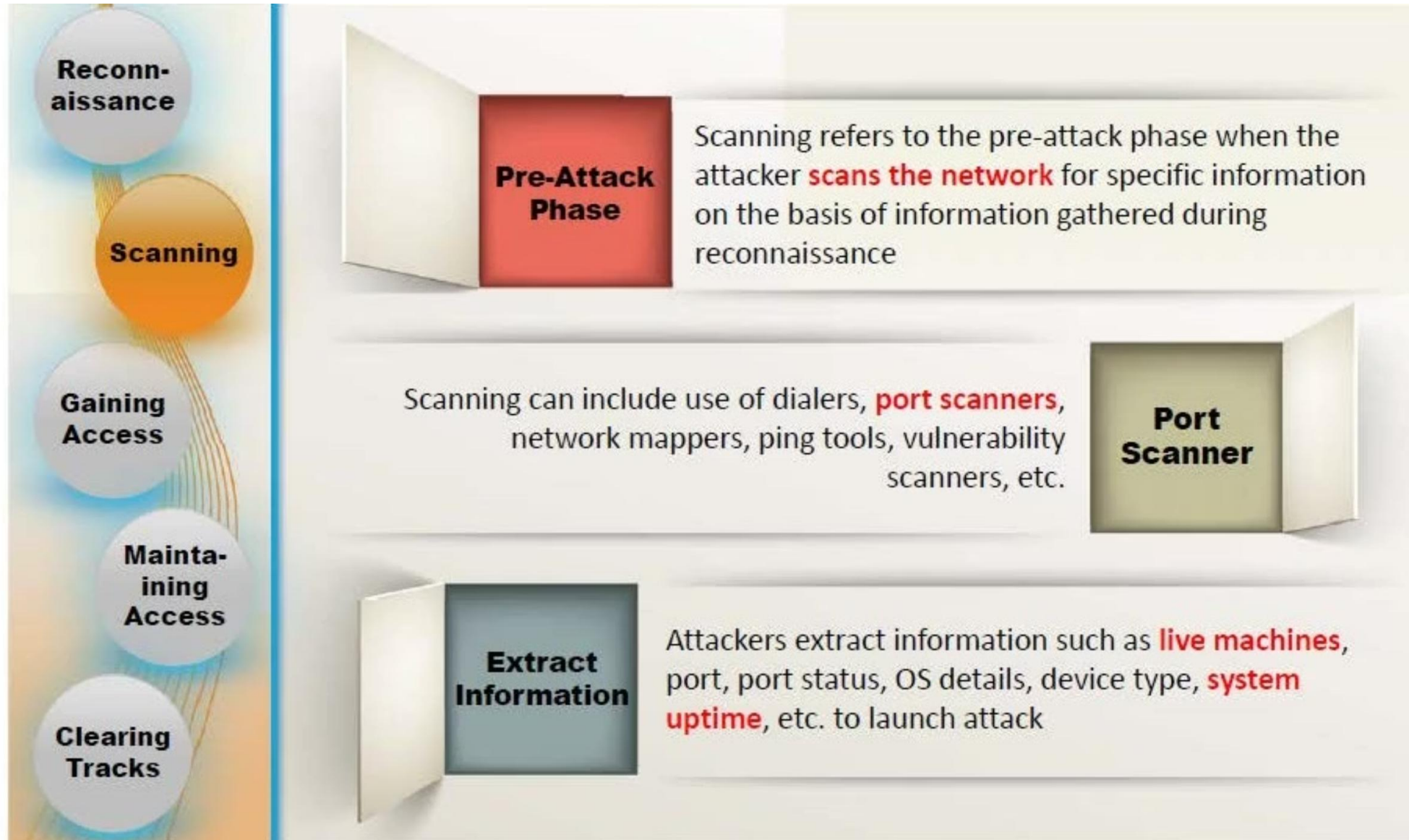
Hacking Terminology

A solid command of hacking terminology is essential for any cybersecurity professional. The following terms are foundational vocabulary used across the field — in industry certifications, vulnerability reports, and security assessments.

Hack Value It is the notion among hackers that something is worth doing or is interesting	Zero-Day Attack An attack that exploits computer application vulnerabilities before the software developer releases a patch for the vulnerability
Vulnerability Existence of a weakness, design, or implementation error that can lead to an unexpected event compromising the security of the system	Daisy Chaining It involves gaining access to one network and/or computer and then using the same information to gain access to multiple networks and computers that contain desirable information
Exploit A breach of IT system security through vulnerabilities	Doxing Publishing personally identifiable information about an individual collected from publicly available databases and social media
Payload Payload is the part of an exploit code that performs the intended malicious action, such as destroying, creating backdoors, and hijacking computer	Bot A “bot” is a software application that can be controlled remotely to execute or automate predefined tasks

- Continuing our review of key hacking terminology.
- Familiarity with these concepts allows security professionals to communicate precisely about threats, attack vectors, and defensive countermeasures in both technical and professional contexts.

Hacking Terminology



- These additional terms cover concepts commonly encountered during security assessments, penetration tests, and incident response activities.
- Understanding this vocabulary is critical for interpreting reports and applying security frameworks in real-world scenarios.

Hacking Terminology (Part 4)

Reconn-
aissance

Scanning

Gaining
Access

Mainta-
ining
Access

Clearing
Tracks

Gaining access refers to the point where the attacker obtains access to the **operating system or applications** on the computer or network



The attacker can gain access at the **operating system level**, **application level**, or **network level**



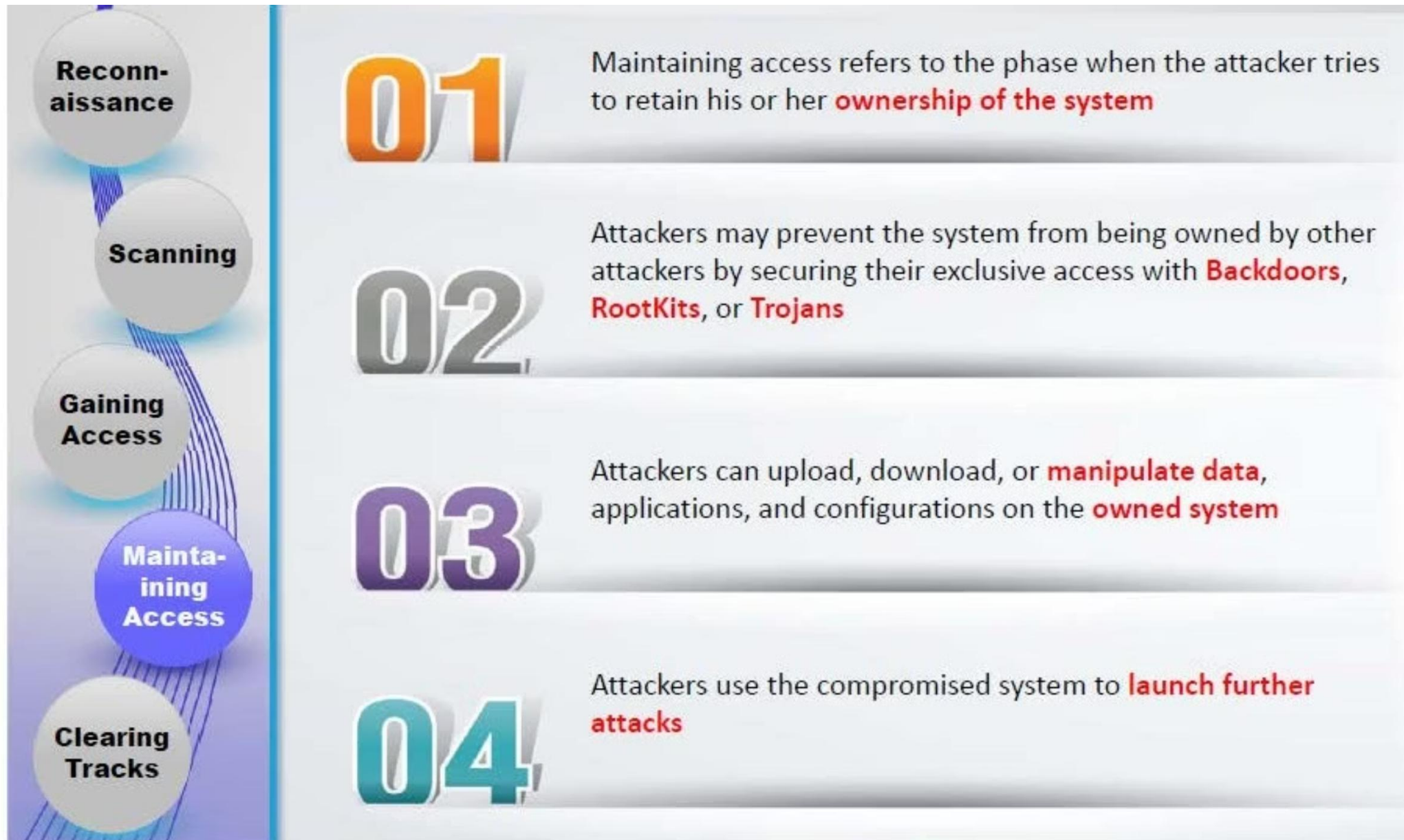
The attacker can **escalate privileges** to obtain complete control of the system. In the process, intermediate systems that are connected to it are also compromised

Examples include **password cracking**, buffer overflows, denial of service, **session hijacking**, etc.

This section introduces further terms related to attack techniques and defensive mechanisms.

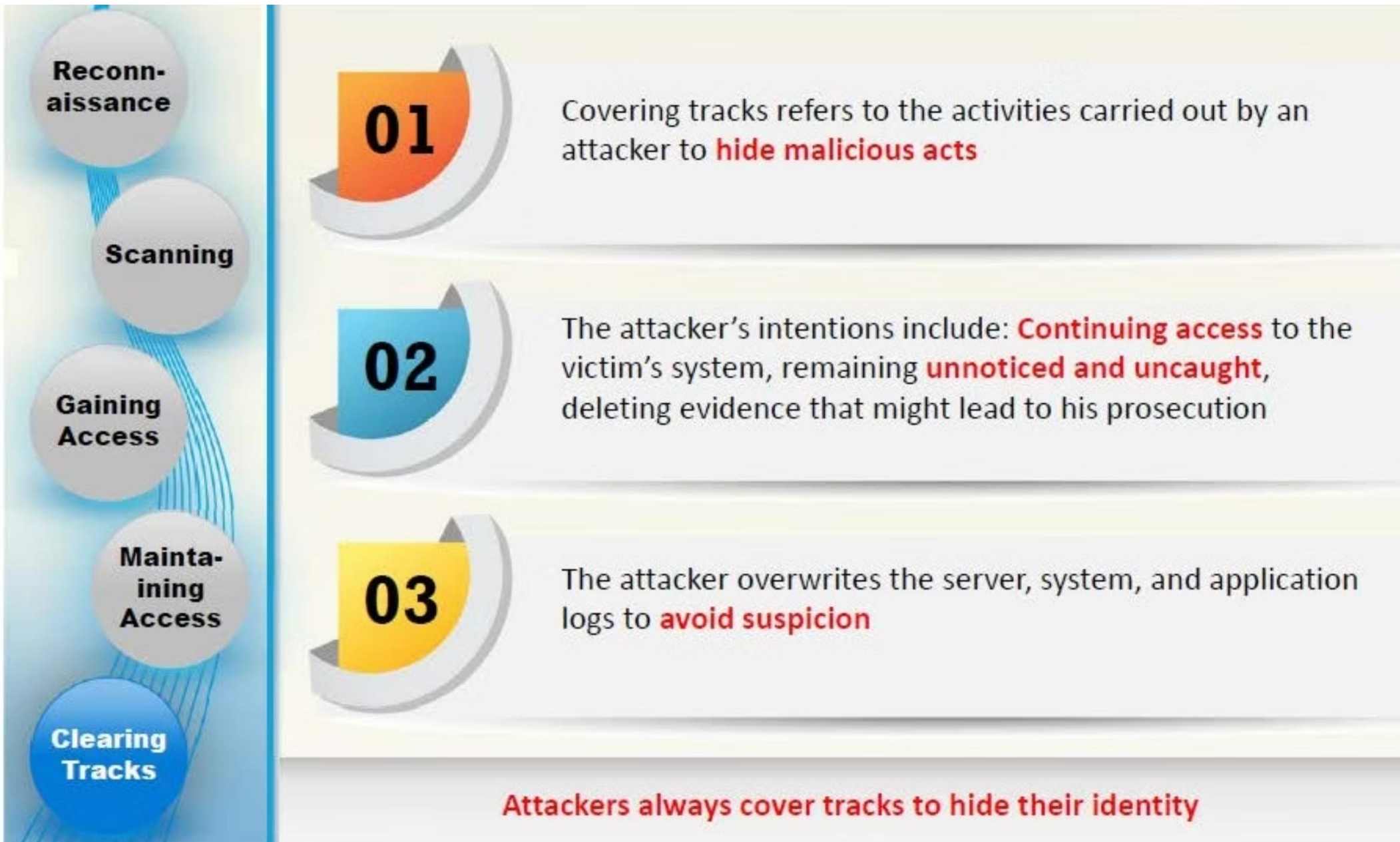
Mastering this vocabulary equips you to engage with advanced cybersecurity literature, professional discussions, and certification examination content with confidence.

Hacking Terminology (Part 5)



- Expanding on our terminology guide, this section addresses concepts relevant to system exploitation and post-compromise activities.
- These terms are especially relevant when analyzing attack chains and understanding the full lifecycle of a cyberattack.

Hacking Terminology (Part 6)



This section covers terminology associated with social engineering, reconnaissance, and information gathering — phases that often precede technical exploitation.

Recognizing these terms helps practitioners build a holistic understanding of the attacker's perspective.

Hacking Terminology (Part 7)

- Incident management is a set of defined processes to **identify, analyze, prioritize, and resolve security incidents** to restore normal service operations as quickly as possible and prevent future recurrence of the incident



- Further key terms relating to network-level attacks and defensive architectures.
- Understanding network-specific terminology is essential for professionals working in areas such as firewall management, intrusion detection, and network security monitoring.

Hacking Terminology (Part 8)



- The final set of core hacking terminology covers concepts related to cryptographic attacks, privilege escalation, and persistence mechanisms.
- These topics are fundamental to both offensive security testing and defensive hardening strategies.

Responsibilities of the Incident Management Team



- When a security incident occurs, the **Incident Management Team (IMT)** plays a pivotal role in coordinating the response.
- Their responsibilities span detection, containment, eradication, recovery, and post-incident analysis — ensuring that organizations can respond swiftly and effectively to minimize damage.

Vulnerability Assessment

A vulnerability assessment is a systematic process of identifying, quantifying, and prioritizing security weaknesses in a system, network, or application. It provides organizations with a clear picture of their current security posture and guides remediation efforts to address the most critical risks first.

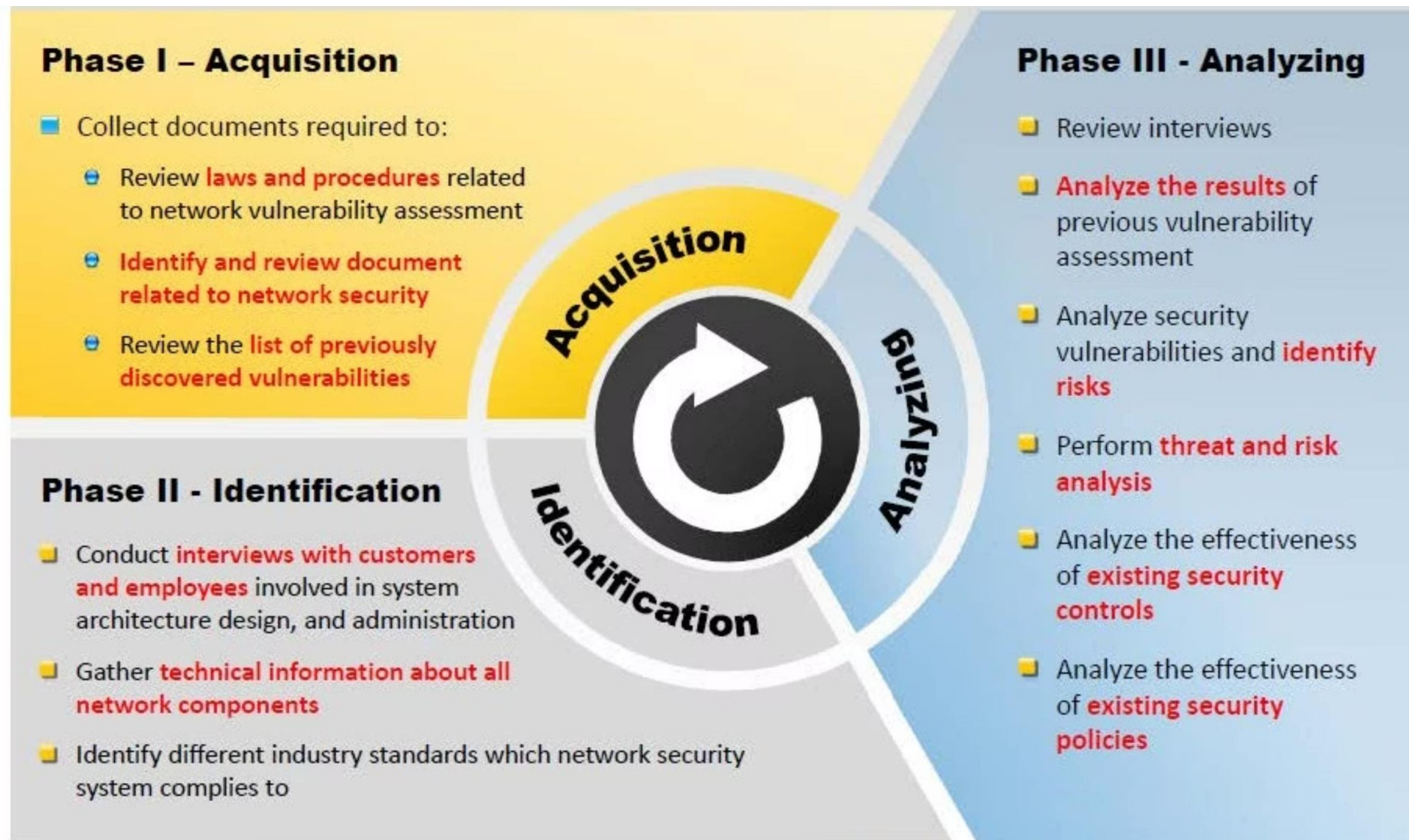
Vulnerability Assessment Types

Vulnerability assessments are not one-size-fits-all. Different environments and objectives call for different assessment approaches. Understanding the available types allows security teams to select the most appropriate methodology for their specific context and risk profile.



Network Vulnerability Assessment Method (Part 1)

Network vulnerability assessments follow a structured methodology to systematically probe network infrastructure for weaknesses. This involves active scanning, passive monitoring, and configuration review to uncover exposed services, misconfigurations, and unpatched systems across the network environment.



Network Vulnerability Assessment Method (Part 2)

Continuing the network vulnerability assessment methodology, this section focuses on the analysis, reporting, and remediation planning phases. Translating raw scan data into prioritized, actionable recommendations is what transforms a vulnerability assessment into genuine organizational value.

Vulnerability Research

Vulnerability research involves the discovery, analysis, and documentation of security weaknesses in software, hardware, and network systems. Security professionals use a combination of automated tools and manual techniques to identify vulnerabilities, often referencing established databases such as the CVE (Common Vulnerabilities and Exposures) system.

- The process of **discovering vulnerabilities and design flaws** that will open an operating system and its applications to attack or misuse
- Vulnerabilities are classified based on **severity level** (low, medium, or high) and **exploit range** (local or remote)



An administrator needs vulnerability research:

To gather information about **security trends, threats, and attacks**

To know **how to recover** from a network attack

To find **weaknesses**, and alert the network administrator before a **network attack**

To **get information** that helps to prevent the security problems

Vulnerability Research Websites

Staying current with known vulnerabilities is a core responsibility for security professionals. The following websites are essential resources for discovering, tracking, and analyzing published vulnerabilities. Regular consultation of these platforms supports proactive patch management and threat intelligence activities.



CodeRed Center
<http://www.eccouncil.org>



Microsoft Vulnerability Research (MSVR)
<http://technet.microsoft.com>



Security Magazine
<http://www.securitymagazine.com>



SecurityFocus
<http://www.securityfocus.com>



Help Net Security
<http://www.net-security.org>



HackerStorm
<http://www.hackerstorm.co.uk>



SC Magazine
<http://www.scmagazine.com>



Computerworld
<http://www.computerworld.com>



HackerJournals
<http://www.hackerjournals.com>



WindowsSecurity
<http://www.windowsecurity.com>

NVD

National Vulnerability Database — NIST's comprehensive CVE repository.

CVE Details

Searchable CVE database with severity scores and affected products.

Exploit-DB

Archive of public exploits and corresponding vulnerable software.

SecurityFocus

Vulnerability information and security community discussion forums.

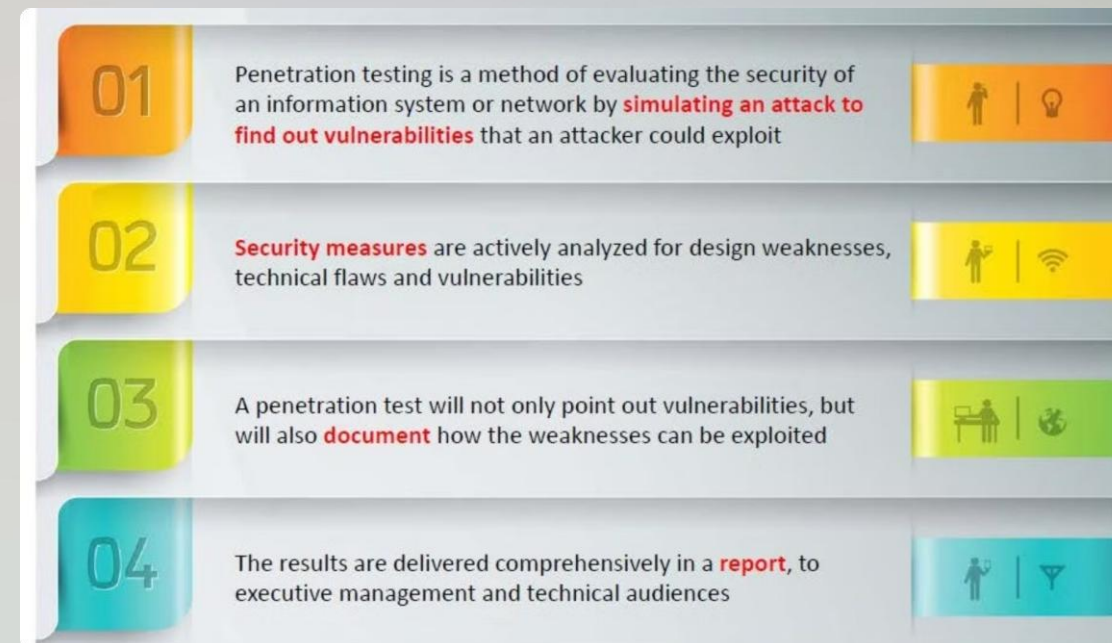


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Penetration Testing: Introduction

Penetration testing is an authorized, simulated cyberattack performed on a computer system, network, or application to evaluate its security. Unlike a vulnerability assessment, penetration testing actively attempts to exploit identified weaknesses to determine the real-world impact of a successful attack. It is one of the most powerful tools in the ethical hacker's arsenal.



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Types of Penetration Testing

Penetration testing is categorized based on the level of information provided to the tester prior to engagement. Each type simulates a different attacker scenario, offering distinct insights into an organization's security resilience.

Black Box

No prior knowledge — simulates an external attacker with no insider information.

White Box

Full knowledge — tester has complete access to system architecture, source code, and credentials.

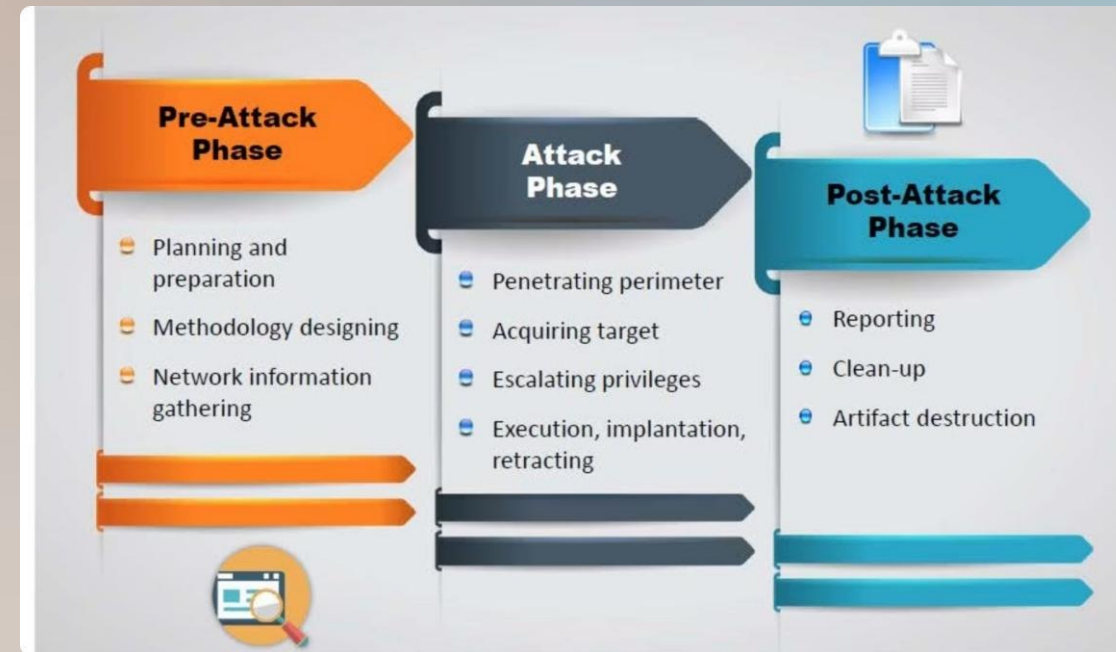
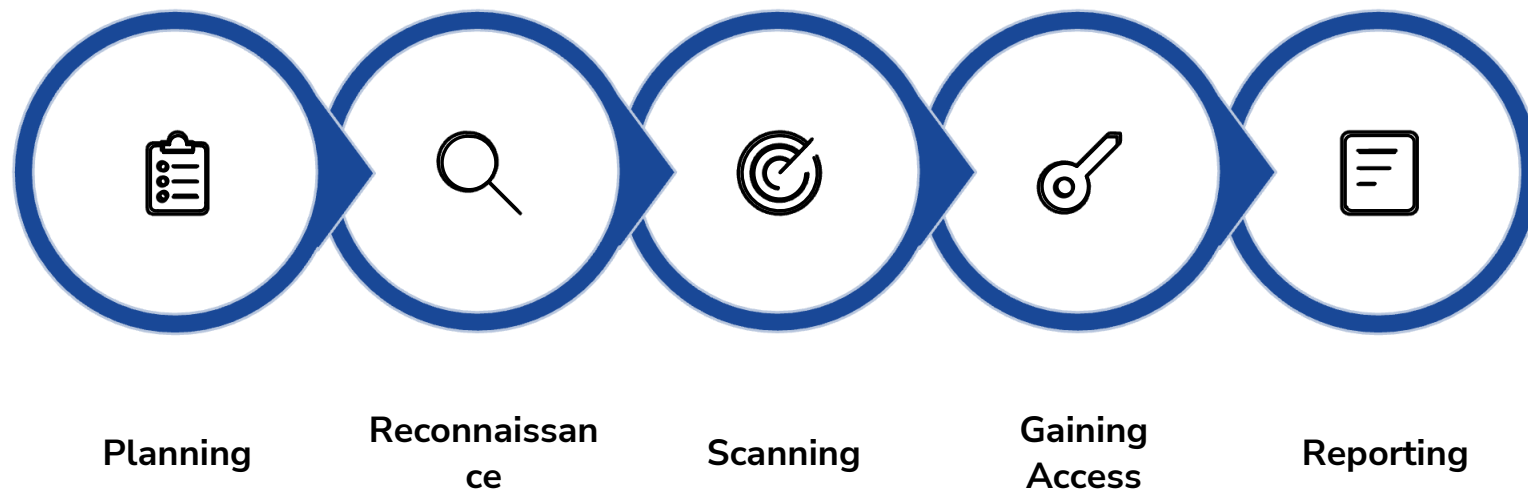
Gray Box

Partial knowledge — simulates an insider threat or attacker with limited access.



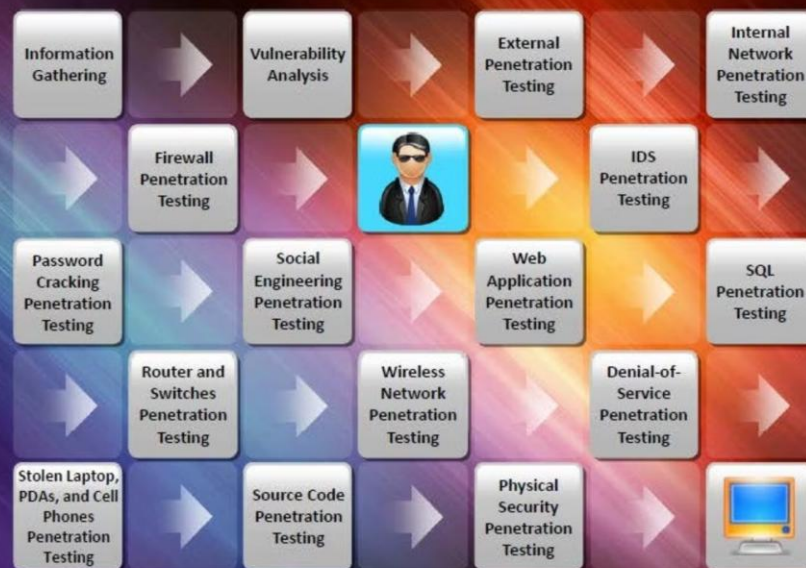
Phases of Penetration Testing

A structured penetration test follows a defined lifecycle that ensures comprehensive coverage, reproducible results, and actionable outcomes. Each phase builds on the previous, guiding the tester from initial scoping through exploitation to final reporting.



Penetration Testing: Tools and Techniques

Ethical hackers utilize a wide range of specialized tools during penetration tests. These tools support activities across all phases — from initial reconnaissance and network scanning to exploitation and post-exploitation. Proficiency with industry-standard tools is a core competency for any penetration tester.



Penetration Testing: Reporting and Deliverables

The penetration testing report is the primary deliverable of any engagement. A well-structured report communicates findings clearly to both technical and non-technical audiences, includes risk-rated vulnerability details, evidence of exploitation, and prioritized remediation recommendations that the organization can act upon immediately.





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Hands-On Task: Submission Instructions

After completing the quiz, take a screenshot of your result. Upload the screenshot to the **Week 1 Hands-On Task** submission box on Moodle:

❏ **Submission Link:** <https://lms.cihe.edu.au/course/view.php?id=831§ion=8>



Next Class: Web Security & Hacking

In our next session, we move from foundational concepts to applied web security. We will examine how attackers target web servers and applications — and how ethical hackers test and secure them.



Web Server & App Penetration Testing

Techniques for testing web servers and applications for known and zero-day vulnerabilities.



Vulnerabilities Testing

Hands-on methods for identifying and validating common web vulnerabilities such as SQL injection and XSS.



Web Application Hacking

Understanding how attackers exploit web applications — and how to replicate these attacks ethically.



Securing Web Environments

Best practices and tools for hardening websites, web applications, and web server configurations.



Thank You

Q & A

Thank you for your attention and engagement throughout this lecture. We have covered the foundations of ethical hacking — from cybercrime categories and legal frameworks to hacker types, vulnerability assessment, and penetration testing methodologies. Please bring any questions to the floor — this is your opportunity to clarify concepts before the next session.

